

Policy Optimization for Corrupted Markov Decision Processes

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Abstract

Policy optimization methods are a widely used and empirically effective class of reinforcement learning algorithms. In this paper, we use policy optimization to solve the challenging problem of efficient learning in *corrupted* Markov Decision Processes (MDPs) with both adversarial losses and transitions. In particular, we develop a corruption-robust policy optimization algorithm, CR-UOB-PO, that achieves $\mathcal{O}(\sqrt{T} + C^P)$ regret in the finite tabular MDP setting under known C^P , which quantifies the total transition corruption across all episodes. We further demonstrate how CR-UOB-PO can be reduced using a principled black-box approach to maintain the same regret bound *without* knowledge of C^P . To our knowledge, this is the first algorithm for corrupted MDPs that relies almost entirely on policy optimization methods, allowing it to handle extreme environments with low computational complexity.

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1 Introduction

Markov Decision Processes (MDPs) have served as the main tool to formalize sequential decision making under uncertainty through state transitions and loss functions, enabling applications across robotics, resource management, and autonomous systems. Within this framework, finite-horizon T -episodic MDPs with adversarial loss functions have been extensively studied, with Jin et al. [2020] and Rosenberg & Mansour [2019] establishing algorithms that achieve $\mathcal{O}(\sqrt{T})$ regret. In contrast, fully adversarial MDPs (adversarial losses and transition functions) pose significantly greater challenges. Studies have shown that with bandit feedback and adversarially corrupted transitions, regret can grow exponentially with episode length Abbasi Yadkori et al. [2013], Tian et al. [2021]. To address this limitation, Jin et al. [2023] introduced an occupancy-measure-based algorithm that smoothly interpolates between the result obtained for fixed transitions adversarial MDPs (where $C^P = 0$) Jin et al. [2020] and for linearly corrupted transitions (where $C^P = \mathcal{O}(T)$) Tian et al. [2021]. Their divergence-based policy achieves an optimal $\mathcal{O}(\sqrt{T} + C^P)$ regret bound, which scales linearly with corruption.

Policy Optimization (PO) methods Agarwal et al. [2020b], Abbasi-Yadkori et al. [2019] represent an alternative approach that has demonstrated remarkable empirical effectiveness in artificial intelligence, robotics, and video gaming applications Schulman et al. [2017a,b]. PO methods offer distinct practical advantages over occupancy-measure-based approaches: they exhibit lower computational complexity, scale more efficiently to large state spaces, and integrate naturally with function approximation techniques [Sun et al., 2024]. However, PO has traditionally been constrained by unrealistic global exploration requirements and has lacked the theoretical guarantees against adversarial environments that occupancy-measure-based algorithms provide [Jin et al., 2023]. Recent advances Cai et al. [2020], Shani et al. [2020b], Agarwal et al. [2020a] have focused on integrating exploration *bonuses* to overcome these exploration constraints. Notably, Luo et al. [2021] developed a PO approach with dilated exploration bonuses that achieves $\mathcal{O}(\sqrt{T})$ regret in MDPs with adversarial losses

and bandit feedback, but still assumes fixed transitions. This naturally leads to our central research question: *can we achieve $\mathcal{O}(\sqrt{T} + C^{\mathsf{P}})$ regret using policy optimization in MDPs with both adversarial losses and transitions while retaining the computational advantages of PO methods?*

Our research provides an affirmative answer to this question by extending the policy optimization framework in a principled manner. Based on Luo et al. [2021], we develop an adapted algorithm for tabular settings with known corruption measure C^{P} (Section 4). At its core, our method combines the confidence set for corrupted transition functions of Jin et al. [2023] with newly designed exploration bonuses, b^C and B^C , which significantly improve robustness against transition corruption (Section 3). This approach allows us to achieve the regret bound of $\tilde{\mathcal{O}}(\sqrt{T} + C^{\mathsf{P}})$ (Section 5), which matches the optimal bounds given in Jin et al. [2023]. We then extend these theoretical guarantees to the more challenging scenario of *unknown corruption* (Section 6), which allows our results to generalize to a wider set of problems and ensures maximal practical viability. To our knowledge, this work represents the first policy optimization method that matches the state-of-the-art regret bounds for adversarial MDPs with corrupted transitions, while maintaining the implementation and computational advantages that have made PO methods the preferred approach in most practical applications.

Related Works Our research connects to several established areas within reinforcement learning, the closest being corruption-robust learning. Lykouris et al. [2023] established the first $\mathcal{O}(\sqrt{T})$ regret guarantee for corrupted transitions with multiplicative dependence on the number of corrupted episodes. Chen et al. [2021] improved upon this with more precise corruption measurements, while Zhang et al. [2021] used policy gradient methods for bounded corruption settings. Wei et al. [2024] established a sum-of-per-round-corruption measure, and Jin et al. [2023] introduced the corruption measure C^{P} used in our paper, achieving $\mathcal{O}(\sqrt{T} + C^{\mathsf{P}})$ regret with divergence-based algorithms.

Another closely related approach considers non-stationary MDPs with systematically changing environments. Cheung et al. [2020] developed confidence-widening techniques for low dynamic regret, Gajane et al. [2018] proposed sliding window approaches, and Wei & Luo [2021] introduced a black-box reduction achieving optimal dynamic regret without prior knowledge of non-stationarity. Similarly, Yu & Mannor [2009] combined online learning with robust control for arbitrarily changing rewards and transitions.

The challenge of handling adversarial elements has also been addressed through various policy optimization approaches. He et al. [2022] achieved near-optimal regret for linear MDPs with adversarial rewards, while Shani et al. [2020b] established sub-linear regret bounds for unknown transitions with bandit feedback. Advancing this line of work, Tiapkin et al. [2025] bridged the gap between adversarial and stochastic MDPs, and Zhong et al. [2024] presented the first provably efficient policy optimization scheme for non-stationary environments.

Beyond adversarial settings, policy optimization techniques have evolved across multiple dimensions. Shani et al. [2020a] analyzed TRPO convergence rates, Liu et al. [2022] improved constrained MDP convergence, and Russel et al. [2020], Dong et al. [2022], and Sun et al. [2024] developed complementary approaches for robust MDPs with model uncertainties.

Our work bridges corruption-robust learning with efficient policy optimization, in order to achieve near-optimal results while still encouraging reduced computational complexity and increased sample efficiency. In particular, we extend Luo et al. [2021]’s dilated bonuses approach to handle corrupted transitions while incorporating Jin et al. [2023]’s Corral extension to unknown C^P , thereby achieving optimal $\tilde{O}(\sqrt{T} + C^P)$ regret even with adversarial losses and transitions.

2 Problem Setting

Preliminaries We consider a finite-horizon episodic MDP similar to Luo et al. [2021] and Jin et al. [2023], characterized by a finite state space S that admits a layered structure (S is

partitioned into subsets $S_0, \dots, S_H$ where $|S_0| = |S_H| = 1$, and transitions are only possible from S_h to S_{h+1} for $h \in [H] := \{0, \dots, H-1\}$, a finite action space A , a stochastic transition function $P_t(\cdot | s, a)$ specifying the distribution of the next state after taking action a in state s , and an adversarial loss function $\ell_t(s, a)$ for every episode $1 \leq t \leq T$. An episode here refers to a trajectory from S_0 to S_H that follows the actions and transitions and suffers the losses defined above.

Our environment is also defined similarly to Jin et al. [2023], where an adversary is allowed to perturb the transitions $\{P_t\}_{t=1}^T$ arbitrarily and with knowledge of the learner's algorithms. The learner then sequentially interacts with the T MDPs, where for each episode t :

1. The learner decides on a stochastic policy $\pi : S \times A \rightarrow [0, 1]$ where $\pi_t(a|s)$ is the probability of taking action a when visiting state s , and (re-)starts their state at $s_{t,0}$.
2. For every $h \in [H]$ with according state $s_{t,h}$, the learner then:
 - (a) selects action $a_{t,h}$ sampled from $\pi_t(s_{t,h}, a_{t,h})$,
 - (b) suffers *adversarial* loss $\ell_t(s_{t,h}, a_{t,h})$,
 - (c) and moves to $s_{t,h+1}$ sampled from $P_t(\cdot | s_{t,h}, a_{t,h})$ (potentially corrupted).
3. Finally, the learner receives bandit feedback of *only* the suffered losses at the end of the episode, and updates the algorithm accordingly.

The extent to which the adversary is allowed to modify the transitions is called *corruption* and is quantified in Jin et al. [2023], re-stated below:

$$C^P = \min_{P' \in \mathcal{P}} \sum_{t=1}^T \sum_{h=0}^{H-1} \max_{(s,a) \in S_h \times A} \|P_t(\cdot | s, a) - P'(\cdot | s, a)\|_1. \quad (1)$$

Abusing notation slightly, we shall use P to denote the uncorrupted optimal transition (i.e. the fixed transition P that achieves C^P). Note that the learner is unable to observe the

amount and episodes that the adversary corrupts the transitions. However, as of now, we assume that the total corruption C^P is given. In section 6, we will use a similar reduction method given in Jin et al. [2023] to give a regret bound when C^P is unknown.

We further define the Bellman equations that will be used extensively below:

$$\begin{cases} V(s_H, \ell) = 0, \\ Q^\pi(s, a, \ell) = \ell(s, a) + \mathbb{E}_{s' \sim P(\cdot|s,a)}[V^\pi(s', \ell)], \\ V^\pi(s, \ell) = \mathbb{E}_{a \sim \pi(\cdot|s)}[Q^\pi(s, a, \ell)], \end{cases} \quad (2)$$

where $V^\pi(s, \ell)$ is the state value function and $Q^\pi(s, a, \ell)$ is the state-action value function. We further overload notation by defining $V_t^\pi(s) := V_t^\pi(s, \ell_t)$ and $Q_t^\pi(s, a) := Q_t^\pi(s, a, \ell_t)$.

The goal of the learner is to minimize regret, defined as

$$R_T = \sum_{t=1}^T V_t^{\pi_t}(s_0) - \min_{\pi} \sum_{t=1}^T V_t^{\pi}(s_0). \quad (3)$$

We also define $\pi^* = \arg \min_{\pi} \sum_{t=1}^T V_t^{\pi}(s_0)$ as an optimal *fixed* policy in hindsight.

Occupancy Measures We define $q^\pi(s)$ as the probability of visiting state s within an episode while following π . We further define $q^\pi(s, a) = q^\pi(s)\pi(s, a)$ as the probability of visiting state s and performing action a within an episode while following π , $q_t(s, a) = q^{\pi_t}(s, a)$ to clarify episode t , and $q^* = q^{\pi^*}$ where π^* is defined above.

By definition, we then have $V^\pi(s_0, \ell) = \sum_{s,a} q^\pi(s, a)\ell(s, a)$, and we will extend ℓ to accommodate any function $b : S \times A \rightarrow \mathbb{R}$ to support our bonus functions in the next section.

Corruption-Robust Confidence Sets A standard technique to deal with unknown transitions in MDPs is to use shrinking confidence sets, which have been used extensively in MDP literature Auer and Ortner [2006], Auer et al. [2008], Rosenberg and Mansour [2019]. However, for handling corrupted transitions specifically, we need to modify this approach.

We utilize the enlarged, corruption-robust confidence set as introduced in Jin et al. [2023]. To summarize their construction, the estimation procedure proceeds in epochs $i = 1, 2, \dots$ of episodes. An epoch ends whenever a state-action pair is encountered whose total number of visits doubles that at the beginning of the epoch. At the beginning of every epoch i , we calculate the empirical transition $\bar{P}_i$ as:

$$\bar{P}_i(s' | s, a) = \frac{m_i(s, a, s')}{m_i(s, a)}, \forall (s, a, s') \in W_h, h \in [H - 1], \quad (4)$$

where $W_h = S_h \times A \times S_{h+1}$ and $m_i(s, a)$ and $m_i(s, a, s')$ are the total number of visits to (s, a) and (s, a, s') respectively prior to epoch i . Using this estimate, we construct the confidence set:

Definition 2.1. Let $\delta \in (0, 1)$ be a confidence parameter. With known corruption C^P , we define the confidence set of transition functions for epoch i as

$$\begin{aligned} \mathcal{P}_i = \{ & P \in \mathcal{P} : |P(s' | s, a) - \bar{P}_i(s' | s, a)| \\ & \leq B_i(s, a, s'), \forall (s, a, s') \in W_h, h \in [H] \} \end{aligned} \quad (5)$$

where the confidence interval $B_i(s, a, s')$ is defined, with $\iota = |S||A|T/\delta$, as

$$\begin{aligned} B_i(s, a, s') = \min \left\{ & 1, 16 \sqrt{\frac{\bar{P}_i(s' | s, a) \log(\iota)}{m_i(s, a)}} \right. \\ & \left. + 64 \cdot \frac{C^P + \log(\iota)}{m_i(s, a)} \right\}. \end{aligned} \quad (6)$$

By enlarging the confidence sets, we are able to ensure with high probability that the true *uncorrupted* transitions will be contained in the transition set. Lemma 2.2 of Jin et al. [2023] quantifies this by proving that $\mathbb{P}(P \in \mathcal{P}_i \forall i) \geq 1 - 2\delta$, the details of which will be analyzed in the Appendix.

3 Dilated Exploration Bonuses and Regret

One of the most important lemmas we will use is the performance difference lemma Kakade and Langford [2002], which allows us to rewrite our regret as follows:

$$R_T = \sum_s \sum_{t=1}^T q_t^*(s) \sum_a (\pi_t(a | s) - \pi^*(a | s)) Q_t^{\pi_t}(s, a)$$

Notice here that q_t^* is indexed by t because even though the best policy in hindsight with respect to the changing probabilities transitions is fixed, the probability of visiting state s changes each episode due to the changing probability transitions. Also note that $Q_t^{\pi_t}(s, a)$ is also affected by the changing transitions at time t (not just the policy π_t).

Recall that in Luo et al. [2021], *dilated exploration bonuses* (b and B) were added to the Q -value functions, which encourage more exploration and make the algorithm more robust to adversarial rewards. However, their bonuses did not account for adversarial transitions, which warrants a new definition of corrupted bonuses. Hence, we define our new b^C and B^C as follows:

$$\left\{ \begin{array}{l} b_t^C(s) = \mathbb{E}_{a \sim \pi_t(\cdot | s)} \left[\frac{3\gamma H + H(\bar{q}_t(s, a) - \underline{q}_t(s, a))}{\bar{q}_t(s, a) + \gamma} \right] \\ \quad + 2|S||A|H \frac{C^P}{T}, \\ B_t^C(s, a) = b_t^C(s) + \left(1 + \frac{1}{H} \right) \\ \quad \cdot \max_{\hat{P} \in \mathcal{P}_i} \mathbb{E}_{s' \sim \hat{P}(\cdot | s, a)} \mathbb{E}_{a' \sim \pi_t(\cdot | s')} [B_t^C(s', a')], \\ B_t^C(s_H, a) = 0 \quad \forall a \in A, \end{array} \right. \quad (7)$$

where $\bar{q}_t(s, a) = \max_{P \in P_k} q^{\hat{P}, \pi_t}(s, a)$ is the upper occupancy measure of q i.e. the maximum value of q in the confidence set, and $\underline{q}_t(s, a) = \min_{P \in P_k} q^{\hat{P}, \pi_t}(s, a)$ is the lower occupancy measure of q i.e. the minimum value of q in the confidence set. Both can be calculated using **COMP-UOB** and **COMP-LOB** respectively Jin et al. [2020], Luo et al. [2021], with **Greedy** as a prerequisite algorithm for these two. Details of these algorithms will be provided in the

Appendix (see Section B).

Notice the added term $2|S||A|HC^P/T$ to b_t^C . Intuitively, one can draw parallels to optimistic UCB error terms in multi-armed bandit settings and consider *per-episode* corruption as an additional bonus for unexplored actions, encouraging more exploration to compensate for corrupted transitions. This formulation also allows us to preserve the recursive form of B_t^C , allowing us to prove our version of Lemma 3.1 in Luo et al. [2021].

Lemma 3.1. (*Extension of Lemma 3.1 in Luo et al. [2021]*) *If*

$$\begin{aligned} \sum_s q_t^*(s) \sum_{t=1}^T \langle \pi_t(\cdot|s) - \pi^*(\cdot|s), Q_t^{\pi_t}(s, \cdot) - B_t^C(s, \cdot) \rangle \\ \leq o(T) + \sum_{t=1}^T V^{\pi^*}(s_0; b_t^C) \\ + \frac{1}{H} \sum_{t=1}^T \sum_{s,a} q_t^*(s) \pi_t(a|s) B_t^C(s, a) \end{aligned}$$

holds, then

$$R_T \leq o(T) + 3 \sum_{t=1}^T V^{\pi_t}(s_0; b_t^C).$$

Proof. Note that our definitions of B_t^C and B_t are similar, so we can apply the same decomposition from Lemma B.1 of Luo et al. [2021] (A.4) to achieve the same result. Details will be deferred to the Appendix. $\square$

Thus, in later sections, we will show it suffices for an algorithm for this problem to achieve regret R_T in the form of Lemma (3.1) with high probability (i.e. $1 - O(\delta)$).

4 The Known Corruption Case

Here we propose a policy optimization algorithm, CR-UOB-PO, that operates under known total corruption C^P . The pseudocode, along with an overall outline, is presented in Algorithm 1.

Algorithm 1 Corruption-Robust Upper Occupance Bound Policy Optimization (CR-UOB-PO)

Parameters: $\delta \in (0, 1)$,

$$\eta = \min \left\{ 1/(24H^3 + 24|A||S|H^4), 1/\sqrt{|S||A|HT} \right\},$$

$$\gamma = 2\eta H.$$

Initialization: Set epoch index $i = 1$ and confidence set $\mathcal{P}_1$ as the set of all transition functions. For all (s, a, s') , initialize counters $m_0(s, a) = m_1(s, a) = 0, m_0(s, a, s') = m_1(s, a, s') = 0$.

for $t = 1, 2, \dots, T$ **do**

Step 1: Compute and execute policy. Execute π_t for one episode, where

$$\pi_t(a | s) \propto \exp \left(-\eta \sum_{\tau=1}^{t-1} \left(\hat{Q}_\tau(s, a) - B_\tau^C(s, a) \right) \right),$$

and obtain trajectory $\{(s_{t,h}, a_{t,h}, \ell_t(s_{t,h}, a_{t,h}))\}_{h=0}^{H-1}$.

Step 2: Construct Q -function estimators. For all $h \in [H]$ and $(s, a) \in S_h \times A$,

$$\hat{Q}_t(s, a) = \frac{L_{t,h}}{\bar{q}_t(s, a) + \gamma} \mathbb{I}_t(s, a),$$

with $L_{t,h} = \sum_{i=h}^{H-1} \ell_t(s_{t,i}, a_{t,i})$, $\bar{q}_t(s, a) = \max_{\hat{P} \in \mathcal{P}_i} q^{\hat{P}, \pi_t}(s, a)$, and $\mathbb{I}_t(s, a) = \mathbb{I}\{s_{t,h} = s, a_{t,h} = a\}$.

Step 3: Construct bonus functions. For all $(s, a) \in S \times A$, define B_t^C and b_t^C as in (7).

Step 4: Update model estimation. $\forall h < H, m_i(s_{t,h}, a_{t,h}) \stackrel{+}{\leftarrow} 1, m_i(s_{t,h}, a_{t,h}, s_{t,h+1}) \stackrel{+}{\leftarrow} 1$.

if $\exists h, m_i(s_{t,h}, a_{t,h}) \geq \max\{1, 2 \cdot m_{i-1}(s_{t,h}, a_{t,h})\}$ **then**

Increment epoch index $i \stackrel{+}{\leftarrow} 1$ and copy counters: $m_i \leftarrow m_{i-1}, m_i \leftarrow m_{i-1}$.

Compute empirical transition $\bar{P}_i(s' | s, a) = \frac{m_i(s, a, s')}{\max\{1, m_i(s, a)\}}$ and confidence set:

$$\begin{aligned} \mathcal{P}_i &= \left\{ \hat{P} : \left| \hat{P}(s' | s, a) - \bar{P}_i(s' | s, a) \right| \right. \\ &\quad \left. \leq B_i(s', a, s), \forall (s, a, s') \in S_h \times A \times S_{h+1} \right\}, \end{aligned}$$

where $B_i(s', a, s)$

$$\begin{aligned} &= \min \left\{ 1, 16 \sqrt{\frac{\bar{P}_i(s' | s, a) \log(|S||A|T/\delta)}{m_i(s, a)}} \right. \\ &\quad \left. + 64 \cdot \frac{C^P + \log(|S||A|T/\delta)}{m_i(s, a)} \right\} \end{aligned}$$

end if
end for

We address finer details of implementation here:

1. We use an enlarged confidence set similar to Jin et al. [2023], which builds on that of Luo et al. [2021] to handle unknown transitions on top of unknown losses. This is to ensure that $\mathbb{P}(P \in \mathcal{P}_i \forall i)$ with high probability (Lemma 2.2. of Jin et al. [2023]).
2. The doubling epoch trick is a well-known trick which dates back to Maurer & Pontil [2009] and Jin et al. [2020]. This allows us to only use $O(|S||A| \log T)$ epochs while only having a $\tilde{O}(L|S|\sqrt{|A|T})$ regret.
3. The optimistic estimators b_t^C , B_t^C and $\hat{Q}_t$ will help guide exploration toward uncertain or potentially corrupted states while maintaining optimal regret bounds. All three quantities can be efficiently computed using **Greedy** (see Algorithm 4 in Appendix), which furthermore solidifies the computational efficiency of our approach.

These details allow us to efficiently execute the entire Algorithm 1, which is crucial for a complete policy optimization design. The regret of Algorithm 1 is shown below:

Theorem 4.1. *CR-UOB-PO gives the following regret bound with probability at least $1 - \tilde{O}(\delta)$:*

$$R_T \leq \tilde{O} \left(H^4 + |A||S|H^5 + |S|H^2\sqrt{|A|T} + |S||A|HC^P \right)$$

The full proof will be deferred to section C. This result is crucial because it achieves the near-optimal regret $\mathcal{O}(\sqrt{T} + C^P)$, matching Jin et al. [2023] while being much more computationally efficient using policy optimization techniques.

5 Corrupted Regret Decomposition

Here, we provide a sketch of our regret analysis and proof of Theorem 4.1 in section C. Recall our goal was achieve regret of the form of Lemma 3.1. To this end, we decompose the

left-hand side of the inequality in Lemma 3.1 into $\text{ERROR}_1 + \text{BIAS}_1 + \text{BIAS}_2 + \text{ERROR}_2 + \text{REG-TERM}$, as shown below:

$$\begin{aligned}
& \sum_{t,h} \mathbb{E}_{s_h \sim \pi^*} [\langle \pi_t(\cdot \mid s_h) - \pi^*(\cdot \mid s_h), Q_t(s_h, \cdot) - B_t^C(s_h, \cdot) \rangle] \\
&= \underbrace{\sum_{t=1}^T \sum_{h=0}^{H-1} \mathbb{E}_{s_h \sim \pi^*} [\langle \pi_t(\cdot \mid s_h), Q_t(s_h, \cdot) - Q_t^P(s_h, \cdot) \rangle]}_{\text{ERROR}_1} \\
&+ \underbrace{\sum_{t=1}^T \sum_{h=0}^{H-1} \mathbb{E}_{s_h \sim \pi^*} [\langle \pi_t(\cdot \mid s_h), Q_t^P(s_h, \cdot) - \widehat{Q}_t(s_h, \cdot) \rangle]}_{\text{BIAS}_1} \\
&+ \underbrace{\sum_{t=1}^T \sum_{h=0}^{H-1} \mathbb{E}_{s_h \sim \pi^*} [\langle \pi^*(\cdot \mid s_h), \widehat{Q}_t(s_h, \cdot) - Q_t^P(s_h, \cdot) \rangle]}_{\text{BIAS}_2} \\
&+ \underbrace{\sum_{t=1}^T \sum_{h=0}^{H-1} \mathbb{E}_{s_h \sim \pi^*} [\langle \pi^*(\cdot \mid s_h), Q_t^P(s_h, \cdot) - Q_t(s_h, \cdot) \rangle]}_{\text{ERROR}_2} \\
&+ \underbrace{\sum_{t=1}^T \sum_{h=0}^{H-1} \mathbb{E}_{s_h \sim \pi^*} [\langle \pi_t(\cdot \mid s_h) - \pi^*(\cdot \mid s_h), \widehat{Q}_t(s_h, \cdot) - B_t^C(s_h, \cdot) \rangle]}_{\text{REG-TERM}}
\end{aligned}$$

where Q_t represents the Q-value function for π_t under actual corrupted transitions from the environment, while Q_t^P denotes the Q-value function for π_t under uncorrupted ground truth transitions.

Our novel approach comes from the terms ERROR_1 and ERROR_2 , which extends from the decomposition in Luo et al. [2021] and measures the performance discrepancy attributed specifically to transition corruption, regardless of which policies $\{\pi_t\}$ are executed. The following lemma addresses both error terms simultaneously, establishing that their combined regret is bounded by $O(ASH^2C^P)$.

Lemma 5.1. *For any sequence of policies $\pi_1, \dots, \pi_T$, we have*

$$\sum_{t,h} \mathbb{E}_{s_h \sim \pi_t} [\langle \pi_t(\cdot | s_h), Q_t(s_h, \cdot) - Q_t^P(s_h, \cdot) \rangle] \leq ASH^2 C^P.$$

For the remaining terms BIAS_1 , BIAS_2 , REG-TERM , note that they operate under the assumption of uncorrupted transitions, and we can use proof techniques close to Jin et al. [2020] and Luo et al. [2021]. We must take care of subtlety with the added bonuses due to corruption, including adjustments to the worst-case learning rate and high probability results. However, the intuition behind the terms remains the same: BIAS_1 captures the amount of underestimation of $\hat{Q}_t$ related to π_t , BIAS_2 captures the amount of overestimation of $\hat{Q}_t$ related to π^* , and REG-TERM is controlled by the multiplicative weight update in our algorithm.

Lemma 5.2. *With probability at least $1 - 3\delta$, BIAS_1 is upper bounded as:*

$$\begin{aligned} & \sum_{t=1}^T \sum_s q^*(s) \langle \pi_t(\cdot | s), Q_t^P(s, \cdot) - \hat{Q}_t(s, \cdot) \rangle \\ & \leq \tilde{\mathcal{O}}\left(\frac{H}{\eta}\right) \\ & + \sum_{t=1}^T \sum_{s,a} q^*(s) \pi_t(a | s) \left(\frac{2\gamma H + H (\bar{q}_t(s, a) - \underline{q}_t(s, a))}{\bar{q}_t(s, a) + \gamma} \right). \end{aligned}$$

Lemma 5.3. *With probability at least $1 - 3\delta$, BIAS_2 is upper bounded as:*

$$\begin{aligned} & \sum_{t=1}^T \sum_{h=0}^{H-1} \sum_s q^*(s) \langle \pi^*(\cdot | s_h), \hat{Q}_t(s_h, \cdot) - Q_t^P(s_h, \cdot) \rangle \\ & \leq \tilde{\mathcal{O}}\left(\frac{H}{\eta}\right). \end{aligned}$$

Lemma 5.4. *With probability at least $1 - 3\delta$, REG-TERM is upper bounded as:*

$$\begin{aligned} & \sum_{t=1}^T \sum_s q^*(s) \left\langle \pi_t(\cdot \mid s) - \pi^*(\cdot \mid s), \widehat{Q}(s, \cdot) - B_t^C(s, \cdot) \right\rangle \\ & \leq \tilde{\mathcal{O}}\left(\frac{H}{\eta}\right) \\ & + \sum_{t=1}^T \sum_{s,a} q^*(s) \pi_t(a \mid s) \left(\frac{\gamma H}{\bar{q}_t(s, a) + \gamma} + \frac{B_t^C(s, a)}{H} \right) \end{aligned}$$

Adding these results gives us Theorem 4.1, and we defer all proofs to Section C in the Appendix.

6 Transitioning to Unknown C^P

We now demonstrate how CR-UOB-PO can be turned into an algorithm that achieves the same regret order when C^P is unknown. The black-box model selection procedure we follow is the same as that outlined in Section 4 of Jin et al. [2023]. In summary, the reduction procedure has two layers:

1. **Bottom Layer (STABILISE):** First, we apply Algorithm 2 from Jin et al. [2023], which transforms any *corruption-robust* algorithm (i.e. CR-UOB-PO) into a *stable* corruption-robust algorithm. The exact definitions are given below. The resulting algorithm still requires known C^P but gains stability properties which are crucial for the second layer.

2. **Top Layer (Corral-Based Selection):** Next, we employ Algorithm 3 from Jin et al. [2023], which builds on the Corral family of model selection approaches Agarwal et al. [2017], Luo et al. [2022]. This layer takes our stable algorithm, sets hypotheses on C^P for each arm at powers of 2, and produces a version that achieves the same regret guarantees without requiring prior knowledge of C^P .

Bottom Layer We start by defining *corruption-robust* algorithms below:

Definition 6.1. (Definition 4.1 of Jin et al. [2023]) An adversarial MDP algorithm is **corruption-robust** if it takes θ (a guess on the corruption amount) as input, and achieves the following regret for any random stopping time $t' \leq T$:

$$\begin{aligned} \max_{\pi} \mathbb{E} \left[\sum_{t=1}^{t'} (V^{\pi_t}(s_0) - V^{\pi}(s_0)) \right] &\leq \mathbb{E} \left[\sqrt{\beta_1 t'} \right. \\ &\left. + (\beta_2 + \beta_3 \theta) \mathbb{I}\{t' \geq 1\} \right] + \Pr[C_{1:t'}^{\pi} > \theta] HT \end{aligned}$$

for problem-dependent constants and $\log(T)$ factors $\beta_1 \geq H^2$, $\beta_2 \geq H$, $\beta_3 \geq 1$, where $C_{1:t'}^{\pi} = \sum_{t=1}^{t'} C_t^{\pi}$ is the total corruption up to time t' .

As explained in Jin et al. [2023], for Definition 6.1, if the guess θ on $C^{\mathbf{P}}$ is larger or equal to the true amount, a corruption-robust algorithm should be of order $\sqrt{t'} + \theta$, which is satisfied by CR-UOB-PO; otherwise the bound is vacuous as it would be $O(HT)$. This ensures that STABILISE can handle adversarial cases efficiently, and we show that CR-UOB-PO indeed satisfies this requirement.

Remark 6.2. CR-UOB-PO satisfies Definition 6.1.

Proof. First, note that extending our algorithm from fixed T to *anytime* (i.e true for all $t' \leq T$) only requires a multiplied constant Besson and Kaufmann [2018], which preserves the regret bounds shown in Section 4. It suffices that we find the proper values for β_i . As $R_T \leq \tilde{O} \left(H^4 + |A||S|H^6 + |S|H^2\sqrt{|A|T} + |S||A|H^2C^{\mathbf{P}} \right)$, we have $\beta_1 = \tilde{O}(|A||S|^2H^4)$, $\beta_2 = \tilde{O}(H^4 + |A||S|H^6)$, and $\beta_3 = \tilde{O}(|A||S|H^2)$. Note that these satisfy $\beta_1 \geq H^2$, $\beta_2 \geq H$, and $\beta_3 \geq 1$, and we're done. $\square$

Now that CR-UOB-PO satisfies Definition 6.1, we can use it as the input for STABILISE, the pseudocode of which can be found in Section B of the Appendix. Before we continue, however, we will define a special learning environment using w_t :

Protocol 1 (from Jin et al. [2023]) *For every round t , before the learner makes a decision, a probability $w_t \in [0, 1]$ is revealed to the learner. After making a decision, the*

learner sees the desired feedback from the environment with probability w_t and sees nothing with probability $1 - w_t$.

This provides interaction between the top and bottom layers and regulates both stability issues (which scales with $1/w_t$) and corruption scaling (which scales with w_t). The general idea is then as follows:

1. The algorithm initializes $\lceil \log_2 T \rceil$ instances of stabilized CR-UOB-PO ($\text{ALG}_1, \text{ALG}_2, \dots, \text{ALG}_{\lceil \log_2 T \rceil}$), where each ALG_j is configured with a scaled corruption parameter $\theta_j = 2^{-j+1}C^P + 16H \log(T)$.
2. In each round t :
 - When receiving probability w_t , STABILISE identifies which instance to use based on the range w_t falls into.
 - If w_t is too small ($\leq \frac{1}{T}$), it ignores the round,
 - Otherwise, it selects instance ALG_{j_t} where $w_t \in (2^{-j_t-1}, 2^{-j_t}]$ and plays the policy suggested by that instance.
3. When feedback is received (which happens with probability w_t), it is only forwarded to ALG_{j_t} with probability $\frac{2^{-j_t-1}}{w_t}$.

This approach ensures that each instance receives feedback with a consistent probability 2^{-j_t-1} , which checks off all concerns shown above, especially by limiting the regret degradation by $\rho_T = \frac{1}{\min_{t \in [T]} w_t} \leq T$. The result is a *stable* corruption-robust algorithm whose regret smoothly degrades with ρ_T and depends linearly on C^P :

Definition 6.3. (Definition 4.2 of Jin et al. [2023]) A $\frac{1}{2}$ -stable corruption-robust algorithm is one that, *with prior knowledge on C^P* , achieves

$$R_T \leq \mathbb{E} \left[\sqrt{\beta_1 \rho_T T} + \beta_2 \rho_T \right] + \beta_3 C^P$$

under *Protocol 6* for problem-dependent constants and $\log(T)$ factors $\beta_1 \geq H^2$, $\beta_2 \geq H$, and $\beta_3 \geq 1$.

By the following theorem from Jin et al. [2023], we can supply CR-UOB-PO to STABILISE and convert it into a $\frac{1}{2}$ -stable corruption-robust algorithm.

Theorem 6.4. (*Theorem 4.3 of Jin et al. [2023]*) *If an algorithm is corruption robust according to 6.1 for some constants $(\beta_1, \beta_2, \beta_3)$, then one can convert it to a $\frac{1}{2}$ -stable corruption-robust algorithm (6.3) with constants $(\beta'_1, \beta'_2, \beta'_3)$ where $\beta'_1 = \mathcal{O}(\beta_1 \log T)$, $\beta'_2 = \mathcal{O}(\beta_2 + \beta_3 L \log T)$, and $\beta'_3 = \mathcal{O}(\beta_3 \log T)$.*

With Theorem 6.4, we can now proceed to the top layer.

Top Layer We utilize the Corral algorithm, which is standard and has been extensively covered in Foster et al. [2021] and Jin et al. [2023]. The general idea is that we utilize $\lceil \log_2 T \rceil$ versions of the $1/2$ -stable corruption robust algorithms with each algorithm having a guess of C^P as a power of 2 (i.e. $C^P \in \{1, 2, 4, \dots, 2^{\lceil \log_2 T \rceil}\}$), as well as another FTRL-based algorithm that chooses the best algorithm on-the-fly among the algorithms above.¹ The pseudocode can be found in Section B; we only state the final regret bound below.

Theorem 6.5. (*Theorem 4.4 of Jin et al. [2023]*) *The resulting algorithms from Definition 6.3, when applied to CORRAL, leads to $R_T = \tilde{\mathcal{O}}(\sqrt{\beta_1 T} + \beta_2 + \beta_3 C^P)$ regret without knowing C^P .*

While we utilized the same approach of Jin et al. [2023], we believe that a change in approach would not lead to a substantial increase in efficiency. This is because the optimization domain and objective function (FTRL with log-barrier regularizer) for CORRAL ($\Delta(M)$) are convex, which implies that we can use highly efficient convex optimization techniques to find w_t . Although this seems like a limitation for computational overhead, and we leave this point

¹While $C^P \leq 2HT$, we consider less CORRAL arms/versions of stabilized algorithms and a lower supremum guess of C^P at $2T$ for lower computational overhead at a negligible regret cost.

for future work, we believe that the theoretical improvements in our regret bounds justify this computational cost, especially as our black-box reduction approach enables more general applications beyond the specific setting studied in this paper.

7 Conclusion

In this paper, we have developed a novel policy optimization approach for learning in corrupted Markov Decision Processes with both adversarial losses and transitions. Building upon the dilated bonus approach of Luo et al. [2021], we introduce a corruption-robust policy optimization algorithm, CR-UOB-PO, which achieves $\tilde{O}(\sqrt{T} + C^P)$ regret bounds while maintaining the computational advantages inherent to policy optimization methods. The key to our approach lies in the carefully re-designed exploration bonuses b^C and B^C , which enable effective learning even when transitions are corrupted adversarially. This represents a significant advance over existing techniques, as our method matches the optimal regret bounds established in Jin et al. [2023] but with substantially reduced computational complexity.

We further extended our results to the more challenging setting where C^P is unknown, maintaining a principled black-box reduction that preserves our optimal regret guarantees. Through this work, we have demonstrated that policy optimization methods can achieve state-of-the-art theoretical guarantees in extremely challenging adversarial environments without sacrificing their practical computational advantages. Future research directions include extending these results to function approximation settings, exploring applications to domains with naturally occurring corruptions, and investigating tighter bounds for specific MDP structures, such as continuous MDPs.

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A Auxiliary Lemmas

Below, we list auxiliary lemmas that will support our analysis. The proofs are available on the respective papers.

Lemma A.1 (A special form of Freedman's inequality, Theorem 1 of Beygelzimer et al. [2011]). *Let $\mathcal{F}_0 \subset \dots \subset \mathcal{F}_n$ be a filtration, and $X_1, \dots, X_n$ be real random variables such that X_i is $\mathcal{F}_i$ -measurable, $\mathbb{E}[X_i|\mathcal{F}_{i-1}] = 0$, $|X_i| \leq b$, and $\sum_{i=1}^n \mathbb{E}[X_i^2|\mathcal{F}_{i-1}] \leq V$ for some fixed $b \geq 0$ and $V > 0$. Then for any $\delta \in (0, 1)$, we have with probability at least $1 - \delta$,*

$$\sum_{i=1}^n X_i \leq \frac{V}{b} + b \log(1/\delta).$$

Throughout the appendix, we let $\mathcal{F}_t$ be the σ -algebra generated by the observations before episode t .

Lemma A.2 (Adapted from Lemma 11 of Jin et al. [2020]). *For all s, a , let $\{z_i(s, a)\}_{i=1}^T$ be a sequence of functions where $z_i(s, a) \in [0, R]$ is $\mathcal{F}_i$ -measurable. Let $Z_i(s, a) \in [0, R]$ be a random variable such that $\mathbb{E}[Z_i(s, a)|\mathcal{F}_i] = z_i(s, a)$. Then with probability at least $1 - \delta$,*

$$\sum_{i=1}^T \sum_{s,a} \left(\frac{1_i(s, a) Z_i(s, a)}{q_i(s, a) + \gamma} - \frac{q_i(s, a) z_i(s, a)}{q_i(s, a)} \right) \leq \frac{RH}{\gamma} \ln \frac{H}{\delta}$$

Lemma A.3 (Regret bound of exponential weight, Theorem 1 of Luo et al. [2021]). *Let $\eta > 0$, and let $\pi_t \in \Delta(A)$ and $\ell_t \in \mathbb{R}^A$ satisfy the following for all $t \in [T]$ and $a \in A$:*

$$\begin{aligned} \pi_1(a) &= \frac{1}{|A|}, \\ \pi_{t+1}(a) &= \frac{\pi_t(a) e^{-\eta \ell_t(a)}}{\sum_{a' \in A} \pi_t(a') e^{-\eta \ell_t(a')}}, \\ |\eta \ell_t(a)| &\leq 1. \end{aligned}$$

Then for any $\pi^ \in \Delta(A)$,*

$$\sum_{t=1}^T \sum_{a \in A} (\pi_t(a) - \pi^*(a)) \ell_t(a) \leq \frac{\ln |A|}{\eta} + \eta \sum_{t=1}^T \sum_{a \in A} \pi_t(a) \ell_t(a)^2.$$

Lemma A.4 (Lemma B.1 of Luo et al. [2021]). *Let $b_t(s, a)$ be a non-negative loss function,*

and $\hat{P}$ be a transition function. Suppose that the following holds for all s, a :

$$\begin{aligned} B_t(s, a) &= b_t(s, a) + \left(1 + \frac{1}{H}\right) \mathbb{E}_{s' \sim \hat{P}(\cdot|s, a)} \mathbb{E}_{a' \sim \pi^*(\cdot|s')} [B_t(s', a')] \\ &\geq b_t(s, a) + \left(1 + \frac{1}{H}\right) \mathbb{E}_{s' \sim P(\cdot|s, a)} \mathbb{E}_{a' \sim \pi^*(\cdot|s')} [B_t(s', a')] \end{aligned}$$

with $B_t(s_H, a) \triangleq 0$, and suppose that the condition in Lemma (3.1) holds. Then

$$\text{Reg} \leq o(T) + 3 \sum_{t=1}^T \hat{V}^{\pi_t}(s_0; b_t).$$

where V^π is the state value function under the transition function $\hat{P}$ and policy π .

Lemma A.5 (Lemma D.3.6 of Jin et al. [2021]). Suppose that a random variable X satisfies the following conditions:

- $X < R$ where R is a constant.
- $X < Y$ conditioning on event A , where $Y \geq 0$ is a random variable.

Then, it holds that $\mathbb{E}[X] \leq \mathbb{E}[Y] + \Pr[A^c] \cdot R$ where A^c is the complementary event of A .

Lemma A.6 (Theorem 5.2.1 of Kakade and Langford [2002]). For any policies π_1, π_2 and any loss function $\ell : S \times A \rightarrow \mathbb{R}$,

$$\begin{aligned} V^{P, \pi_1}(s_0; \ell) - V^{P, \pi_2}(s_0; \ell) &= \sum_{s \neq s_L} \sum_{a \in A} q^{P, \pi_2}(s, a) (V^{P, \pi_1}(s; \ell) - Q^{P, \pi_1}(s, a; \ell)) \\ &= \sum_{s \neq s_L} \sum_{a \in A} q^{P, \pi_2}(s) (\pi_1(a|s) - \pi_2(a|s)) Q^{P, \pi_1}(s, a; \ell). \end{aligned}$$

B Implementation Details for Known and Unknown Corruption

B.1 Known Corruption Algorithms

For completeness, we provide here the algorithms COMP-UOB, COMP-LOB, and GREEDY in Luo et al. [2021] and Jin et al. [2020] for computing the upper and lower occupancy bounds

necessary for the implementation of Algorithm 1, CR-UOB-PO. The bonuses we define in Equation 7, as well as $\hat{Q}_t(s, a)$, can also be computed using GREEDY or a similar procedure. For more details, refer to Luo et al. [2021].

Algorithm 2 COMP-UOB (Algorithm 3 of Jin et al. [2020], Algorithm 4 of Luo et al. [2021])

Require: A policy π_t , a state-action pair (s, a) , and a confidence set $\mathcal{P}$ of the form

$$\left\{ \hat{P} : \left| \hat{P}(s'|s, a) - \bar{P}(s'|s, a) \right| \leq \epsilon(s'|s, a), \forall (s, a, s') \right\}$$

- 1: **Initialize:** for all $\tilde{s} \in S_{h(s)}$, set $f(\tilde{s}) = \mathbb{I}\{\tilde{s} = s\}$
- 2: **for** $h = h(s) - 1$ to 0 **do**
- 3: **for all** $\tilde{s} \in S_h$ **do**
- 4: Compute $f(\tilde{s})$ based on:

$$f(\tilde{s}) = \sum_{a \in A} \pi_t(a|\tilde{s}) \cdot \text{GREEDY}(f, \bar{P}(\cdot|\tilde{s}, a), \epsilon(\cdot|\tilde{s}, a), \max)$$

- 5: **end for**
 - 6: **end for**
 - 7: **return** $\pi_t(a|s)f(s_0)$
-

B.2 Unknown Corruption Algorithms

We also provide the pseudocode for both STABILISE and CORRAL used in Jin et al. [2023] that allow CR-UOB-PO to be turned into an algorithm that operates under unknown C^P . The high-level idea of the Corral selection procedure is to run a collection of base algorithms, each with a different hypothesis on C^P , and run a multi-armed bandit algorithm on top of them so that the algorithm performs almost as well as the best base algorithm. Interested readers can refer to Jin et al. [2023] for proofs of 6.4 and 6.5.

Algorithm 3 COMP-LOB (Algorithm 5 of Luo et al. [2021])

Require: A policy π_t , a state-action pair (s, a) , and a confidence set $\mathcal{P}$ of the form

$$\left\{ \hat{P} : \left| \hat{P}(s'|s, a) - \bar{P}(s'|s, a) \right| \leq \epsilon(s'|s, a), \forall (s, a, s') \right\}$$

- 1: **Initialize:** for all $\tilde{s} \in S_{h(s)}$, set $f(\tilde{s}) = \mathbb{I}\{\tilde{s} = s\}$
- 2: **for** $k = k(s) - 1$ **to** 0 **do**
- 3: **for all** $\tilde{s} \in S_h$ **do**
- 4: Compute $f(\tilde{s})$ based on:

$$f(\tilde{s}) = \sum_{a \in A} \pi_t(a|\tilde{s}) \cdot \text{GREEDY}(f, \bar{P}(\cdot|\tilde{s}, a), \epsilon(\cdot|\tilde{s}, a), \min)$$

- 5: **end for**
 - 6: **end for**
 - 7: **return** $\pi_t(a|s)f(s_0)$
-

Algorithm 4 GREEDY (Algorithm 6 of Luo et al. [2021])

Require: $f : S \rightarrow [0, 1]$, a distribution $\bar{p}$ over n states of layer h , positive numbers $\{\epsilon(s)\}_{s \in S_h}$, objective OPTIMIZE (MAX for COMP-UOB and MIN for COMP-LOB)

- 1: **Initialize:** $j^- = 1$, $j^+ = n$, sort $\{f(s)\}_{s \in S_h}$ and find σ such that
 - 2: **if** OPTIMIZE = max **then**
 - 3: $f(\sigma(1)) \leq f(\sigma(2)) \leq \dots \leq f(\sigma(n))$
 - 4: **else**
 - 5: $f(\sigma(1)) \geq f(\sigma(2)) \geq \dots \geq f(\sigma(n))$
 - 6: **end if**
 - 7: **while** $j^- < j^+$ **do**
 - 8: $s^- = \sigma(j^-)$, $s^+ = \sigma(j^+)$
 - 9: $\delta = \min\{\bar{p}(s^-), \epsilon(s^-)\}$
 - 10: $\delta^+ = \min\{1 - \bar{p}(s^+), \epsilon(s^+)\}$
 - 11: $\bar{p}(s^-) \leftarrow \bar{p}(s^-) - \min\{\delta, \delta^+\}$
 - 12: $\bar{p}(s^+) \leftarrow \bar{p}(s^+) + \min\{\delta, \delta^+\}$
 - 13: **if** $\delta < \delta^+$ **then**
 - 14: $\epsilon(s^+) \leftarrow \epsilon(s^+) - \delta$
 - 15: $j^- \leftarrow j^- + 1$
 - 16: **else**
 - 17: $\epsilon(s^-) \leftarrow \epsilon(s^-) - \delta^+$
 - 18: $j^+ \leftarrow j^+ - 1$
 - 19: **end if**
 - 20: **end while**
 - 21: **return** $\sum_{j=1}^n \bar{p}(\sigma(j))f(\sigma(j))$
-

Algorithm 5 STABLE ALGORITHM BY INDEPENDENT LEARNERS AND INSTANCE SELECTION (STABILISE) (Algorithm 2 of Jin et al. [2023])

Require: C^P and a base algorithm satisfying Definition 6.1.

- 1: **Initialize:** $\lceil \log_2 T \rceil$ instances of the base algorithm $\text{ALG}_1, \dots, \text{ALG}_{\lceil \log_2 T \rceil}$, where ALG_j is configured with the parameter

$$\theta = \theta_j \triangleq 2^{-j+1}C^P + 16H \log(T)$$

- 2: **for** $t = 1, 2, \dots$ **do**
 - 3: Receive w_t
 - 4: **if** $w_t \leq \frac{1}{T}$ **then**
 - 5: play an arbitrary policy π_t
 - 6: **continue** {without updating any instances}
 - 7: **end if**
 - 8: Let j_t be such that $w_t \in (2^{-j_t-1}, 2^{-j_t}]$
 - 9: Let π_t be the policy suggested by ALG_{j_t}
 - 10: Output π_t
 - 11: **if** feedback is received **then**
 - 12: send it to ALG_{j_t} with probability $\frac{2^{-j_t-1}}{w_t}$, and discard it otherwise
 - 13: **end if**
 - 14: **end for**
-

C Regret Proof for Known Corruption (Theorem 4.1)

The full decomposition of the left-hand side of the inequality in Lemma 3.1 is shown here once again for the sake of accessibility:

$$\begin{aligned}
& \sum_{t,h} \mathbb{E}_{s_h \sim \pi^*} [\langle \pi_t(\cdot \mid s_h) - \pi^*(\cdot \mid s_h), Q_t(s_h, \cdot) - B_t^C(s_h, \cdot) \rangle] \\
&= \underbrace{\sum_{t=1}^T \sum_{h=0}^{H-1} \mathbb{E}_{s_h \sim \pi^*} [\langle \pi_t(\cdot \mid s_h), Q_t(s_h, \cdot) - Q_t^P(s_h, \cdot) \rangle]}_{\text{ERROR}_1} \\
&+ \underbrace{\sum_{t=1}^T \sum_{h=0}^{H-1} \mathbb{E}_{s_h \sim \pi^*} [\langle \pi_t(\cdot \mid s_h), Q_t^P(s_h, \cdot) - \hat{Q}_t(s_h, \cdot) \rangle]}_{\text{BIAS}_1} \\
&+ \underbrace{\sum_{t=1}^T \sum_{h=0}^{H-1} \mathbb{E}_{s_h \sim \pi^*} [\langle \pi^*(\cdot \mid s_h), \hat{Q}_t(s_h, \cdot) - Q_t^P(s_h, \cdot) \rangle]}_{\text{BIAS}_2} \\
&+ \underbrace{\sum_{t=1}^T \sum_{h=0}^{H-1} \mathbb{E}_{s_h \sim \pi^*} [\langle \pi^*(\cdot \mid s_h), Q_t^P(s_h, \cdot) - Q_t(s_h, \cdot) \rangle]}_{\text{ERROR}_2} \\
&+ \underbrace{\sum_{t=1}^T \sum_{h=0}^{H-1} \mathbb{E}_{s_h \sim \pi^*} [\langle \pi_t(\cdot \mid s_h) - \pi^*(\cdot \mid s_h), \hat{Q}_t(s_h, \cdot) - B_t^C(s_h, \cdot) \rangle]}_{\text{REG-TERM}}
\end{aligned}$$

Algorithm 6 (A Variant of) CORRAL (Algorithm 3 from Jin et al. [2023])

- 1: **Initialize:** A log-barrier algorithm with each arm being an instance of any base algorithm satisfying Definition 6.1. The hypothesis on C^P is set to 2^i for arm i ($i = 1, 2, \dots, M \triangleq \lceil \log_2 T \rceil$).
- 2: **Initialize:** $\rho_{0,i} = M, \forall i$
- 3: **for** $t = 1, 2, \dots, T$ **do**
- 4: Let

$$w_t = \underset{\substack{w \in \Delta(M) \\ w_i \geq 1/T, \forall i}}{\operatorname{argmin}} \left\{ \left\langle w, \sum_{\tau=1}^{t-1} (\hat{c}_\tau - r_\tau) \right\rangle + \frac{1}{\eta} \sum_{i=1}^M \log \frac{1}{w_i} \right\}.$$
- 5: where $\eta = \frac{1}{4(\sqrt{\beta_1 T} + \beta_2)}$.
- 6: For all i , send $w_{t,i}$ to instance i .
- 7: Draw $i_t \sim w_t$.
- 8: Execute the π_t output by instance i_t .
- 9: Receive the loss c_{t,i_t} for policy π_t (whose expectation is $\ell_t(\pi_t)$) and send it to instance i_t .
- 10: Define for all i :

$$\begin{aligned} \hat{c}_{t,i} &= \frac{c_{t,i} \mathbb{I}[i_t = i]}{w_{t,i}}, \quad \rho_{t,i} = \min_{\tau \leq t} \frac{1}{w_{\tau,i}}, \\ r_{t,i} &= \sqrt{\beta_1 T} (\sqrt{\rho_{t,i}} - \sqrt{\rho_{t-1,i}}) + \beta_2 (\rho_{t,i} - \rho_{t-1,i}). \end{aligned}$$

11: **end for**

Before we show regret proofs, we need to show that the confidence interval is valid:

Lemma C.1 (Lemma 2.2 of Jin et al. [2023]). *Event $\mathcal{E}_{\text{CON}}$ occurs with probability at least $1 - \delta$ where*

$$\mathcal{E}_{\text{CON}} := \left\{ \forall (s, a, s') \in W_h, \forall i \in [T], \forall h < H : \right. \\ \left. |P(s' | s, a) - \bar{P}_i(s' | s, a)| \leq B_i(s, a, s') \right\}.$$

and $B_i(s, a, s')$ is defined as (with $\iota = |S||A|T/\delta$):

$$B_i(s, a, s') \\ = \min \left\{ 1, 16 \sqrt{\frac{\bar{P}_i(s' | s, a) \log(\iota)}{m_i(s, a)}} + 64 \frac{(C^P + \log(\iota))}{m_i(s, a)} \right\}.$$

The proof is in Jin et al. [2023]; we have verified that the proof also applies to our version of the algorithm. For the sake of clarity, this implies $\mathbb{P}(P \in \mathcal{P}_i \forall i) \geq 1 - 2\delta$.

We also cite the corollary, which serves as a general bound on C^P :

Corollary C.2. *Consider any epoch i and any transition $P' \in \mathcal{P}_i$. The following holds:*

$$\|P'(\cdot | s, a) - \bar{P}_i(\cdot | s, a)\|_1 \leq 2 \min \left\{ 1, \frac{32C^P}{\hat{m}_i(s, a)} \right. \\ \left. + 8 \sqrt{\frac{|S_{h(s)+1}| \log \iota}{\hat{m}_i(s, a)}} + \frac{32|S_{h(s)+1}| \log \iota}{\hat{m}_i(s, a)} \right\}.$$

where $\hat{m}_i(s, a) = \max\{m_i(s, a), C^P + \log(\iota)\}$.

With the Lemma and Corollary above, we can now ensure that Step 4 of Algorithm 1 holds. It suffices to show regret bounds of all of the terms shown above.

The following lemma takes care of both ERROR_1 and ERROR_2 at the same time.

Lemma C.3. *For any sequence of policies $[\pi_t]$, we have*

$$\sum_{t=1}^T \sum_{h=0}^{H-1} \mathbb{E}_{s_h \sim \pi_t} [\langle \pi_t(\cdot | s_h), Q_t(s_h, \cdot) - Q_t^P(s_h, \cdot) \rangle] \\ \leq ASH^2 C^P$$

Proof of Lemma 5.1. For the sake of clarity, let

$$Y_h := |P_t(s_{h+1} | s_h, a_h) - P(s_{h+1} | s_h, a_h)|.$$

We first prove the following by backwards induction on the step/layer:

$$\begin{aligned}
& Q_t(s_{h+1}, a_{h+1}) - Q_t^P(s_{h+1}, a_{h+1}) \\
& \leq H \sum_{h'=h+2}^{H-2} \sum_{a_{h'} \in \mathcal{A}_{h'}} \sum_{s_{h'+1} \in S_{h'+1}} |Y_{h'}|.
\end{aligned}$$

Starting from $h = H - 2$ we have as our base case:

$$\begin{aligned}
& Q_t(s_{H-2}, a_{H-2}) - Q_t^P(s_{H-2}, a_{H-2}) \\
& \leq \sum_{a_{H-2} \in \mathcal{A}_{H-2}} \sum_{s_{H-1} \in S_{H-1}} Y_{H-2} \ell(s_{H-1}, a_{H-1}) \\
& \leq H \sum_{a_{H-2} \in \mathcal{A}_{H-2}} \left(\sum_{s_{H-1} \in S_{H-1}} |Y_{H-2}| \right),
\end{aligned}$$

where we have $\ell(s_{H-1}, a_{H-1}) \leq 1 \leq H$.

For the inductive step, suppose the statement holds for $h+1$; we now prove it holds for h .

$$\begin{aligned}
& Q_t(s_h, a_h) - Q_t^P(s_h, a_h) \\
&= \sum_{s_{h+1} \in S_{h+1}} P_t(s_{h+1} | s_h, a_h) \mathbb{E}_{a_{h+1} \sim \pi(\cdot | s_{h+1})} Q_t(s_{h+1}, a_{h+1}) \\
&\quad - P(s_{h+1} | s_h, a_h) \mathbb{E}_{a_{h+1} \sim \pi(\cdot | s_{h+1})} Q_t^P(s_{h+1}, a_{h+1}) \\
&\leq \sum_{s_{h+1} \in S_{h+1}} P_t(s_{h+1} | s_h, a_h) \\
&\quad \cdot \mathbb{E}_{a_{h+1}} \left[Q_t^P(s_{h+1}, a_{h+1}) + H \sum_{h'=h+2}^{H-2} \sum_{a_{h'} \in \mathcal{A}_{h'}} \sum_{s_{h'+1} \in S_{h'+1}} |Y_{h'}| \right] \\
&\quad - P(s_{h+1} | s_h, a_h) \mathbb{E}_{a_{h+1} \sim \pi(\cdot | s_{h+1})} Q_t^P(s_{h+1}, a_{h+1}) \\
&= \sum_{s_{h+1} \in S_{h+1}} \left[Y_h \right] \mathbb{E}_{a_{h+1} \sim \pi(\cdot | s_{h+1})} [Q_t^P(s_{h+1}, a_{h+1})] \\
&\quad + P_t(s_{h+1} | s_h, a_h) \\
&\quad \cdot \mathbb{E}_{a_{h+1} \sim \pi(\cdot | s_{h+1})} \left[H \sum_{h'=h+2}^{H-2} \sum_{a_{h'} \in \mathcal{A}_{h'}} \sum_{s_{h'+1} \in S_{h'+1}} |Y_{h'}| \right] \\
&\leq H \sum_{s_{h+1} \in S_{h+1}} |Y_h| + H \sum_{h'=h+2}^{H-2} \sum_{a_{h'} \in \mathcal{A}_{h'}} \sum_{s_{h'+1} \in S_{h'+1}} |Y_{h'}| \\
&= H \sum_{h'=h+1}^{H-2} \sum_{a_{h'} \in \mathcal{A}_{h'}} \sum_{s_{h'+1} \in S_{h'+1}} |Y_{h'}|,
\end{aligned}$$

where in the first inequality we use the inductive hypothesis above, and in the last inequality $\mathbb{E}_{a_{h+1} \sim \pi(\cdot | s_{h+1})} [Q_t^P(s_{h+1}, a_{h+1})] \leq H$.

This bound on $Q_t - Q_t^P$ now allows us to show that for any policy π_t ,

$$\begin{aligned}
& \sum_{t=1}^T \sum_{h=0}^{H-1} \mathbb{E}_{s_h \sim \pi_t} [\langle \pi_t(\cdot | s_h), Q_t(s_h, \cdot) - Q_t^P(s_h, \cdot) \rangle] \\
& \leq \sum_{t=1}^T \sum_{h=0}^{H-1} H \sum_{h'=h+1}^{H-2} \sum_{a_{h'} \in \mathcal{A}_{h'}} \left(\sum_{s_{h'+1} \in S_{h'+1}} |P_t(s_{h'+1} | s_{h'}, a_{h'}) \right. \\
& \quad \left. - P(s_{h'+1} | s_{h'}, a_{h'}) \right| \Big) \\
& \leq \sum_{t=1}^T \sum_{h=0}^{H-1} H \\
& \quad \cdot \sum_{h'=h+1}^{H-2} \left(|A||S| \max_{(s,a) \in S_h \times A} \|P_t(\cdot | s, a) - P'(\cdot | s, a)\|_1 \right) \\
& \leq |A||S|H \underbrace{\sum_{h=0}^{H-1} \sum_{h'=h+1}^{H-2} \sum_{t=1}^T \max_{(s,a) \in S_h \times A} \|P_t(\cdot | s, a) - P'(\cdot | s, a)\|_1}_{\leq C^P} \\
& = |A||S|H^2 C^P.
\end{aligned}$$

□

For the remaining terms BIAS_1 , BIAS_2 , REG-TERM , we use proof techniques nearly identical to Jin et al. [2020]. Recall we now operate under the assumption of uncorrupted transitions. To this end, we define $q_t^P(s, a)$ (and $q_t^P(s)$) to be the probability of visiting state s and taking action a (and probability of visiting state s , respectively) under the *uncorrupted* transitions following policy π_t .

Lemma C.4. *With probability at least $1 - 3\delta$, BIAS_1 is upper bounded as:*

$$\begin{aligned}
& \sum_{t=1}^T \sum_s q^*(s) \left\langle \pi_t(\cdot | s), Q_t^P(s, \cdot) - \widehat{Q}_t(s, \cdot) \right\rangle \\
& \leq \widetilde{\mathcal{O}} \left(\frac{H}{\eta} \right) \\
& \quad + \sum_{t=1}^T \sum_{s,a} q^*(s) \pi_t(a | s) \left(\frac{2\gamma H + H \left(\bar{q}_t(s, a) - \underline{q}_t(s, a) \right)}{\bar{q}_t(s, a) + \gamma} \right)
\end{aligned} \tag{8}$$

Proof of Lemma 5.2. We operate under the event $\mathcal{E}$ that $P \in \mathcal{P}_i$ (the true uncorrupted transitions) as defined in 2.1 for all epoch i , which holds with probability at least $1 - 2\delta$ by

Lemma 2.2 of Jin et al. [2023]. Under this event, $\underline{q}_t(s, a) \leq q_t^P(s, a) \leq \bar{q}_t(s, a)$ for all t, s, a .

Let $Y_t = \sum_s q^*(s) \langle \pi_t(\cdot | s), \hat{Q}_t(s, \cdot) \rangle$. First, we decompose BIAS-1 as

$$\text{BIAS}_1 = \sum_{t=1}^T \left(\left(\sum_s q^*(s) \langle \pi_t(\cdot | s), Q_t^P(s, \cdot) \rangle \right) - Y_t \right) \quad (9)$$

$$= \sum_{t=1}^T (\mathbb{E}_t[Y_t] - Y_t) \quad (10)$$

$$+ \left(\sum_s q^*(s) \langle \pi_t(\cdot | s), Q_t^P(s, \cdot) \rangle - \mathbb{E}_t[Y_t] \right) \quad (11)$$

We will bound the first term above (which is a Martingale sequence even under the *corrupted setting*) using A.1. By Lemma C.1 of Luo et al. [2021] (adapted for our new transition confidence set), we have the following bound (with probability at least $1 - 2\delta$):

$$\mathbb{E}_t[(\mathbb{E}_t[Y_t] - Y_t)^2] \leq \sum_{s,a} q^*(s) \pi_t(a | s) \frac{H^3}{\bar{q}_t(s, a) + \gamma}$$

Also note that $|Y_t| \leq H \sup_{s,a} |\hat{Q}(s, a)| \leq \frac{H^2}{\gamma}$. Thus, applying A.1, we have that with probability at least $1 - 3\delta$,

$$\begin{aligned} & \sum_{t=1}^T (\mathbb{E}_t[Y_t] - Y_t) \\ & \leq \frac{\gamma}{H^2} \left(\sum_{t=1}^T \sum_{s,a} q^*(s) \pi_t(a | s) \frac{H^3}{\bar{q}_t(s, a) + \gamma} \right) + \frac{H^2}{\gamma} \ln \frac{1}{\delta} \end{aligned}$$

Moreover, by Luo et al. [2021], under $\mathcal{E}$, for every t , the second term in (11) can be bounded as

$$\begin{aligned} & \sum_s q^*(s) \langle \pi_t(\cdot | s), Q_t^P(s, \cdot) \rangle - \mathbb{E}_t[Y_t] \\ & \leq \sum_{s,a} q^*(s) \pi_t(a | s) H \left(\frac{\bar{q}_t(s, a) - \underline{q}_t(s, a) + \gamma}{\bar{q}_t(s, a) + \gamma} \right) \end{aligned}$$

Combining them, we have that with probability at least $1 - 3\delta$,

$$\begin{aligned}
\text{BIAS}_1 &\leq \sum_{t=1}^T \sum_{s,a} q^*(s) \pi_t(a | s) H \left(\frac{(\bar{q}_t(s, a) - \underline{q}_t(s, a)) + \gamma}{\bar{q}_t(s, a) + \gamma} \right) \\
&+ \frac{\gamma}{H^2} \left(\sum_{t=1}^T \sum_{s,a} q^*(s) \pi_t(a | s) \frac{H^3}{\bar{q}_t(s, a) + \gamma} \right) \\
&+ \frac{H^2}{\gamma} \ln \frac{1}{\delta} \\
&\leq \tilde{\mathcal{O}} \left(\frac{H}{\eta} \right) \\
&+ \sum_{t=1}^T \sum_{s,a} q^*(s) \pi_t(a | s) \left(\frac{2\gamma H + H (\bar{q}_t(s, a) - \underline{q}_t(s, a))}{\bar{q}_t(s, a) + \gamma} \right)
\end{aligned}$$

□

Lemma C.5. *With probability at least $1 - 3\delta$, BIAS_2 is upper bounded as:*

$$\sum_{t=1}^T \sum_{h=0}^{H-1} \sum_s q^*(s) \left\langle \pi^*(\cdot | s_h), \hat{Q}_t(s_h, \cdot) - Q_t^P(s_h, \cdot) \right\rangle \leq \tilde{\mathcal{O}} \left(\frac{H}{\eta} \right)$$

Proof of Lemma 5.3. By Lemma C.2 from Luo et al. [2021] (as Q_t^P is under the assumption of uncorrupted transitions), we get that with probability at least $1 - \delta$,

$$\begin{aligned}
&\sum_{t=1}^T \sum_{h=0}^{H-1} \sum_{s,a} q^*(s_h) \pi^*(a | s_h) \left(\hat{Q}_t(s_h, a) - \frac{q_t^P(s_h, a)}{\bar{q}_t(s_h, a)} Q_t^P(s_h, a) \right) \\
&\leq \frac{H^2}{2\gamma} \ln \frac{H}{\delta}
\end{aligned}$$

Note that $\bar{q}_t(s, a) = \max_{\hat{P} \in \mathcal{P}_i} q^{\hat{P}, \pi^t}(s, a)$, is using enlarged confidence set. By Lemma 2.2 of Jin et al. [2023], the true corrupted transition function lies within our confidence set as defined in 2.1 with probability at least $1 - 2\delta$, meaning $q_t^P(s, a) \leq \bar{q}_t(s, a)$ for all t, s, a . Thus, with probability at least $1 - 3\delta$,

$$\begin{aligned}
\text{BIAS}_2 &= \sum_{t=1}^T \sum_{h=0}^{H-1} \sum_{s,a} q^*(s_h) \pi^*(s_h, a) \left(\hat{Q}_t(s_h, a) - Q_t^P(s_h, a) \right) \\
&\leq \frac{H^2}{2\gamma} \ln \frac{H}{\delta} \leq \tilde{\mathcal{O}} \left(\frac{H}{\eta} \right).
\end{aligned}$$

as $\gamma = 2\eta H$. □

Before we prove the REG-TERM bound, we need an auxiliary lemma:

Lemma C.6. $\eta\hat{Q}_t(s, a) \leq \frac{1}{2}$ and $\eta B_t^C(s, a) \leq \frac{1}{2H}$.

Proof of Lemma C.6. Recall that $\gamma = 2\eta H$ and $\eta \leq \frac{1}{24H^3 + 24|A||S|H^4} = \frac{1}{6H^2} \times \frac{1}{4H(1+|A||S|H)}$. Thus,

$$\begin{aligned} \eta\hat{Q}_t(s, a) &\leq \frac{\eta H}{\gamma} = \frac{\eta H}{2\eta H} = \frac{1}{2}, \\ \eta b_t^C(s, a) &= \frac{3\gamma\eta H + \eta H(\bar{q}_t(s, a) - q_t(s, a))}{\bar{q}_t(s, a) + \gamma} + \eta \cdot \left(2|A||S|H \frac{C^P}{T}\right) \\ &\leq 3\eta H + \eta H + 2\eta|A||S|H \times 2H \quad (C^P \leq 2HT) \\ &= 4\eta H(1 + |A||S|H) \\ &\leq \frac{1}{6H^2}. \end{aligned}$$

By the definition of $B_t^C(s, a)$ in 7, we have

$$\eta B_t^C(s, a) \leq H \left(1 + \frac{1}{H}\right)^H \sup_{s', a'} \eta b_t^C(s', a') \leq 3H \times \frac{1}{6H^2} = \frac{1}{2H}$$

This finishes the proof. □

Lemma C.7. *With probability at least $1 - 3\delta$, REG-TERM is upper bounded as:*

$$\begin{aligned} &\sum_{t=1}^T \sum_s q^*(s) \left\langle \pi_t(\cdot | s) - \pi^*(\cdot | s), \hat{Q}(s, \cdot) - B_t^C(s, \cdot) \right\rangle \\ &\leq \tilde{O}\left(\frac{H}{\eta}\right) \\ &+ \sum_{t=1}^T \sum_{s, a} q^*(s) \pi_t(a | s) \left(\frac{\gamma H}{\bar{q}_t(s, a) + \gamma} + \frac{B_t^C(s, a)}{H} \right) \end{aligned}$$

Proof of Lemma 5.4. As mentioned above, we operate under the event $P \in \mathcal{P}_i$, where with probability at least $1 - 2\delta$, $\underline{q}_t(s, a) \leq q_t^P(s, a) \leq \bar{q}_t(s, a)$ for all t, s, a (based on the enlarged corruption set).

By Lemma C.6, we have $\eta|\hat{Q}_t - B_t^C| \leq |\frac{1}{2} - \frac{1}{2H}| \leq 1$. Hence, by Lemma A.4 of Luo et al. [2021], for every state s ,

$$\begin{aligned} & \sum_{t=1}^T \left\langle \pi_t(\cdot | s) - \pi^*(\cdot | s), \hat{Q}_t(s, \cdot) - B_t^C(s, \cdot) \right\rangle \\ & \leq \frac{\ln|A|}{\eta} + \eta \sum_{t=1}^T \sum_{a \in A} \pi_t(a | s) \left(\hat{Q}_t(s, a) - B_t^C(s, a) \right)^2. \end{aligned}$$

Summing over states s with weights $q^*(s)$, we have

$$\begin{aligned} & \text{REG-TERM} \\ & \leq \frac{H \ln|A|}{\eta} + \eta \sum_{t=1}^T \sum_{s,a} q^*(s) \pi_t(a | s) \left(\hat{Q}_t(s, a) - B_t^C(s, a) \right)^2 \\ & \leq \frac{H \ln|A|}{\eta} + 2\eta \sum_{t=1}^T \sum_{s,a} q^*(s) \pi_t(a | s) \hat{Q}_t(s, a)^2 \\ & \quad + 2\eta \sum_{t=1}^T \sum_{s,a} q^*(s) \pi_t(a | s) B_t^C(s, a)^2. \end{aligned}$$

We proceed by analyzing the latter two terms. Note that for the first term,

$$\begin{aligned} & 2\eta \sum_{t=1}^T \sum_{s,a} q^*(s) \pi_t(a | s) \hat{Q}_t(s, a)^2 \\ & \leq 2\eta \sum_{t=1}^T \sum_{s,a} q^*(s) \pi_t(a | s) \frac{H^2 \mathbb{I}_t(s, a)}{(\bar{q}_t(s, a) + \gamma)^2} \\ & = 2\eta H^2 \sum_{t=1}^T \sum_{s,a} \frac{q^*(s) \pi_t(a | s)}{\bar{q}_t(s, a) + \gamma} \cdot \frac{\mathbb{I}_t(s, a)}{\bar{q}_t(s, a) + \gamma} \\ & \leq 2\eta H^2 \sum_{t=1}^T \sum_{s,a} \frac{q^*(s) \pi_t(a | s)}{\bar{q}_t(s, a) + \gamma} \cdot \frac{q_t^P(s, a)}{\bar{q}_t(s, a)} + 2\eta H^2 \times \frac{\frac{H}{\gamma} \ln \frac{H}{\delta}}{2\gamma} \\ & \leq \frac{H}{4\eta} \ln \frac{H}{\delta} + \sum_{t=1}^T \sum_{s,a} q^*(s) \pi_t(a | s) \frac{\gamma H}{\bar{q}_t(s, a) + \gamma}, \end{aligned}$$

where the third line holds with probability $1 - \delta$ by Lemma A.2² of Luo et al. [2021] with setting $z_t(s, a) = Z_t(s, a) = \frac{q^*(s) \pi_t(a | s)}{\bar{q}_t(s, a) + \gamma} \leq \frac{1}{\gamma}$, and the last step uses $\gamma = 2\eta H$ and $q_t^P(s, a) \leq \bar{q}_t(s, a)$, the latter happening with probability $1 - 2\delta$. For the second term, note

²Note here that even though $\hat{Q}$ can be affected by corruption, we still have $L_{t,h} \in [0, H]$.

that

$$2\eta \sum_{t=1}^T \sum_{a \in A} \pi_t(a | s) B_t^C(s, a)^2 \leq \frac{1}{H} \sum_{t=1}^T \sum_{a \in A} \pi_t(a | s) B_t^C(s, a)$$

by Lemma C.4 of Luo et al. [2021]. Combining everything finishes the proof. $\square$

We can now prove Theorem 4.1.

Theorem C.8. CR-UOB-PO gives the following regret bound with probability at least $1 - \tilde{\mathcal{O}}(\delta)$:

$$R_T \leq \tilde{\mathcal{O}} \left(H^4 + |A||S|H^5 + |S|H^2\sqrt{|A|T} + |S||A|HC^P \right)$$

Proof. Putting BIAS₁, BIAS₂, REG-TERM, ERROR₁, and ERROR₂ together, we obtain a bound of the form in Lemma (3.1) that holds with probability at least $1 - \mathcal{O}(\delta)$:

$$\begin{aligned} & \sum_{t=1}^T \sum_{h=0}^{H-1} \mathbb{E}_{s_h \sim \pi^*} [\langle \pi_t(\cdot | s_h) - \pi^*(\cdot | s_h), Q_t(s_h, \cdot) - B_t^C(s_h, \cdot) \rangle] \\ & \leq 2|A||S|H^2C^P + \tilde{\mathcal{O}} \left(\frac{H}{\eta} \right) \\ & + \sum_{t=1}^T \sum_{s,a} q^*(s) \pi^*(a|s) b_t(s, a) + \frac{1}{H} \sum_{t=1}^T \sum_{s,a} q^*(s) \pi_t(a|s) B_t^C(s, a) \\ & = \tilde{\mathcal{O}} \left(\frac{H}{\eta} \right) + \sum_{t=1}^T \sum_{s,a} q^*(s) \pi^*(a|s) \underbrace{\left(b_t(s, a) + 2|A||S|H \frac{C^P}{T} \right)}_{b_t^C(s,a)} \\ & + \frac{1}{H} \sum_{t=1}^T \sum_{s,a} q^*(s) \pi_t(a|s) B_t^C(s, a) \\ & = \tilde{\mathcal{O}} \left(\frac{H}{\eta} \right) + \sum_{t=1}^T \sum_{s,a} q^*(s) \pi^*(a|s) b_t^C(s, a) \\ & + \frac{1}{H} \sum_{t=1}^T \sum_{s,a} q^*(s) \pi_t(a|s) B_t^C(s, a) \end{aligned}$$

where in the first equality we used $\sum_{s,a} q^*(s) \pi^*(a|s) = H$. This bound is now in the form of 3.1. Due to the definition of $B_t^C(s, a)$ in 7 using the optimistic transition function $\hat{P}_t$, the condition for Lemma 3.1 also holds with probability at least $1 - \mathcal{O}(\delta)$ for all t, s, a . Thus

applying Lemma 3.1, we get the bound:

$$\begin{aligned}
R_T &= \tilde{O} \left(\frac{H}{\eta} + \sum_{t=1}^T \sum_{s,a} q^{\hat{P}_t, \pi_t}(s, a) b_t^C(s, a) \right) \\
&= \tilde{O} \left(\frac{H}{\eta} + \sum_{t=1}^T \sum_{s,a} q^{\hat{P}_t, \pi_t}(s, a) \right. \\
&\quad \cdot \left(\frac{3\gamma H + H(\bar{q}_t(s, a) - \underline{q}_t(s, a))}{\bar{q}_t(s, a) + \gamma} + 2|S||A|H \frac{C^P}{T} \right) \Bigg) \\
&= \tilde{O} \left(\frac{H}{\eta} + |S||A|H C^P \right. \\
&\quad \left. + \sum_{t=1}^T \sum_{s,a} q^{\hat{P}_t, \pi_t}(s, a) \left(\frac{3\gamma H + H(\bar{q}_t(s, a) - \underline{q}_t(s, a))}{\bar{q}_t(s, a) + \gamma} \right) \right)
\end{aligned}$$

Through Theorem C.5 of Luo et al. [2021], the last term is bounded by $\tilde{O}(|S|H^2\sqrt{|A|T} + \eta|S||A|H^2T)$, giving us:

$$R_T \leq \tilde{O} \left(\frac{H}{\eta} + |S|H^2\sqrt{|A|T} + \eta|S||A|H^2T + |S||A|H C^P \right)$$

Finally, acknowledging $\eta = \min \left\{ 1/(24H^3 + 24|A||S|H^4), 1/\sqrt{|S||A|HT} \right\}$, we get:

$$R_T \leq \tilde{O} \left(H^4 + |A||S|H^5 + |S|H^2\sqrt{|A|T} + |S||A|H C^P \right)$$

□